# Kevin Shen

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## Education

**University of Waterloo** Sep 2025 - Present
Candidate for BASc in Mechatronics Engineering, GPA: 3.8/4.0

## Experience

**Engineering Co-op** – Multimatic – East Gwillimbury, ON May 2026 – Aug 2026
- Drove root-cause investigations for 8+ PLC and automated system failures by tracing live logic
- Restored industrial weld-cell robots to operation by homing and troubleshooting faults
- Troubleshot industrial sensors by comparing physical activation with PLC inputs using manufacturer data sheets
- Developed PLC ladder-logic changes, creating rungs and tags to improve troubleshooting and maintenance time by ~15%
- Controlled process conditions across 20 stages by interpreting analytical results and adjusting automated equipment

**CAD Drafting Assistant** – Delph & Jenkins Land Surveyors – Aurora, ON Feb 2024 – Jun 2024
- Drafted 10+ subdivision, grading, and property survey plans in AutoCAD for review and submission
- Converted survey measurements, reference plans, and legal descriptions into accurate AutoCAD geometry, validating CAD designs across 100+ survey records

## Projects & Design Teams

**Electric Go-Kart Chassis Design & Fabrication** – Mechanical Design, Welding Aug 2026
- Designed a go-kart chassis in SolidWorks and built the full-scale frame from steel tubing and welded the structure from the CAD design
- Mounted and connected the rack-and-pinion steering system, including the steering wheel, segmented column, universal joints, adjustable tie rods, and front spindles

**Waterloo Rocketry Oxidizer Tank Tower** – Structural Analysis, FMEA Jan 2026 – Jun 2026
- Modeled a preliminary steel oxidizer tank tower in SolidWorks; launched preliminary design review with approved project budget of $5000
- Evaluated critical structural load cases including beam deflection, weld loading, and trailer-mounting loads, performed FMEA and proposed max-load, motion, and vibration validation testing

**Projector Guided Assembly System** – Mechanical Design, 3D Printing, Prototyping Sep 2026
- Designed and built a 2-axis camera/projector mechanism for assembly projection system
- Developed 3D-printed servo couplings, bearing supports, and bolted mounting interfaces for the camera/projector assembly

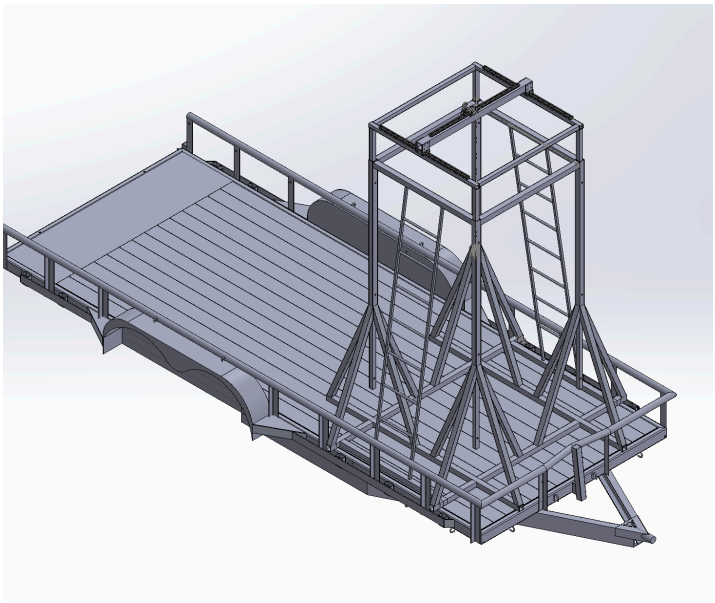
## Skills

**CAD:** SolidWorks, Fusion 360, Onshape, AutoCAD, Drafting and GD&T
**Manufacturing**: Lathe, Mill, MIG-Welding, 3D-Printing
**Controls:** RSLogix 5000, Ladder Logic, FactoryTalk View, PLC/HMI Troubleshooting, PID Controllers
**Electronics:** STM32, ESP32, Arduino, FreeRTOS, CAN, KiCad, PCB Design, Soldering
**Programming:** C, C++, Python, Java, MATLAB, Git

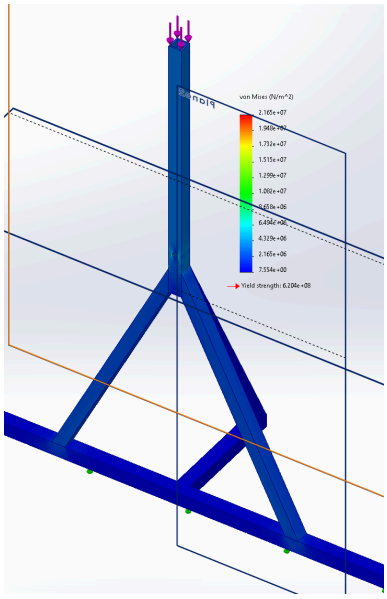
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## Trailer-Mounted Oxidizer Tank Support Tower – Waterloo Rocketry



CAD Rendering of Tank Tower Mounted on Trailer



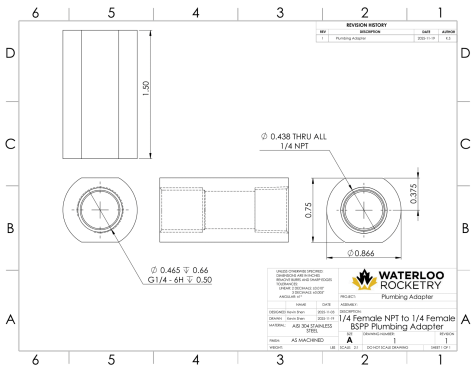
Stress Test Load Simulation

- Designed and modeled a preliminary steel oxidizer tank tower for Waterloo Rocketry, developing the concept through PDR with an approved project budget of $5000
- Defined structural requirements for a 500 kg static load capacity, providing a factor of safety greater than 6 for an 80 kg propellant load while maintaining a freely suspended tank for accurate load-cell measurements
- Developed a telescoping gantry architecture with lockable X, Y, and Z adjustment, combining structural steel tubing, guide-rail carriages, and a hoist system to align the oxidizer tank with feed-system plumbing
- Evaluated critical structural load cases including beam deflection, weld loading, and trailer-mounting loads, performed FMEA and proposed max-load, motion, and vibration validation testing
- Preliminary Design Review presentation available here

## Plumbing Adapter – Waterloo Rocketry



Machined Adapter with Thread




Technical CAD Drawing

- Designed a 1/4” BSPP female to 1/4” NPT female stainless steel adapter for fuel fluid systems; contributed to world record amateur liquid rocket
- Used SolidWorks to create proper drawings and tolerances, then machined the part on lathe and mill

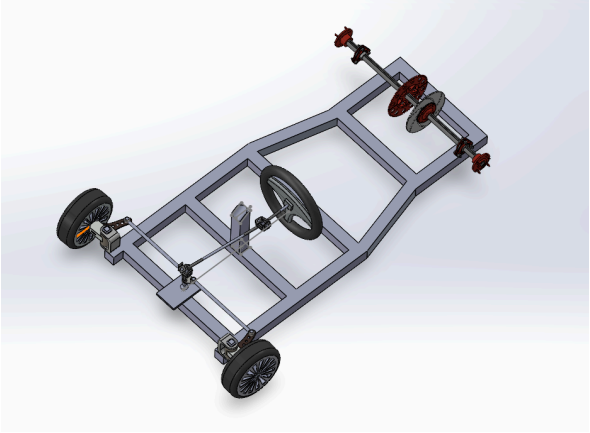
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## Electric Go-Kart Chassis Design & Fabrication



Fabricated Chassis with Installed Steering System



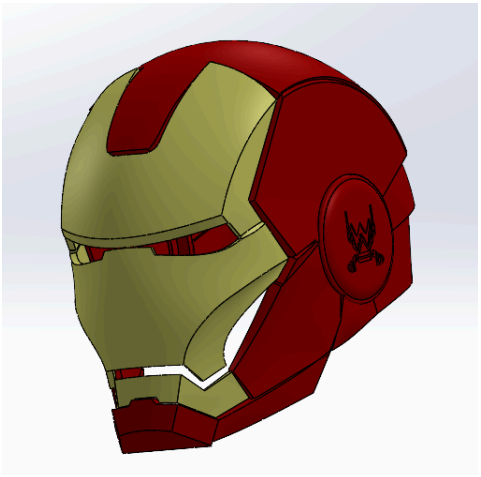
SolidWorks Chassis & Component Assembly

- Designed the go-kart chassis in SolidWorks and built the full-scale frame from steel tubing and welded the structure from the CAD design
- Mounted and connected the rack-and-pinion steering system, including the steering wheel, segmented column, universal joints, adjustable tie rods, and front spindles
- Next steps include adding rear axle and wheels, implementing the drivetrain, and developing autonomous driving

## Iron Man Helmet Prototype - WATBotics



3D Printed and Assembled Helmet



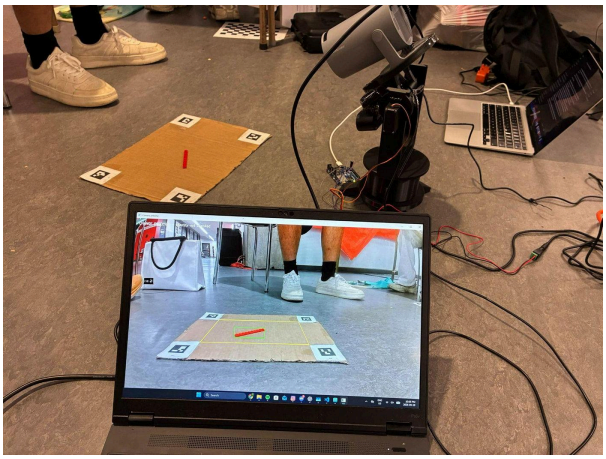
CAD Rendering of Helmet Assembly

- Co-developed a full-scale Iron Man helmet prototype in SolidWorks with a 3D-printed internal frame and removable cosmetic shell
- Led the internal packaging design by defining component locations, wiring paths, mounting space, and hardware clearances within the helmet
- Evaluated 5 candidate polymers for the frame, shell, and visor based on toughness, heat resistance, surface finish, and printability
- Contributed to 3D printing and final assembly, checking the fit between the structural frame, cosmetic shell, and internal hardware

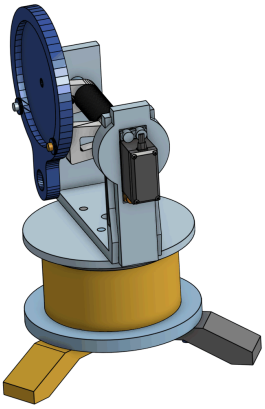
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## Projector-Guided Assembly System – Hack the North 2026



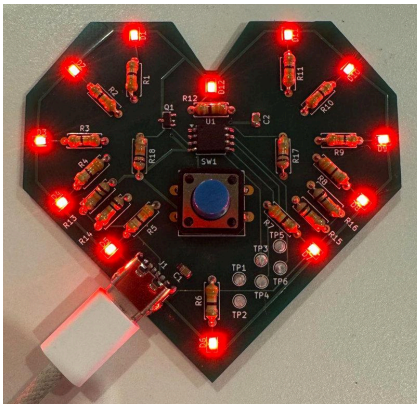
Assembled Camera/Projector Mechanism with Live Object



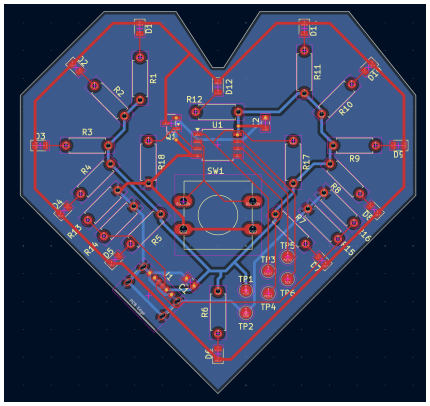
CAD Rendering of Pan-Tilt Structure

- Co-developed a CAD-guided assembly assistant combining OpenCV placement verification with projected cues that identify which component to select and where to place it
- Designed and built a two-axis camera/projector mechanism with 3D-printed servo couplings, bearing supports, and bolted interfaces, integrating a teammate's mounting plate and tilt shaft
- Programmed Arduino C++ firmware to control two MG996R servos through a PCA9685 over I2C, with independent PWM limits and home positions
- Implemented USB serial commands for axis positioning, status reporting, and output disable to support hardware setup and integration
- Devpost available here and code available here

## Programmable Heart LED PCB



Assembled PCB



KiCAD PCB Layout

- Designed and assembled a custom 12-LED heart-shaped PCB in KiCad, integrating 5 V USB-C power, an ATtiny85, and MOSFET-switched PWM control
- Programmed and debugged embedded C++ firmware to illuminate 12 LEDs at power-on and execute a button-triggered PWM sequence with debounced input, timed pulses, and rapid fade-to-dim transitions
- Hand-assembled and debugged the PCB, isolating solder bridges and connection faults through continuity testing and reworking the board with flux and solder wick
- Integrated current-limiting resistors, decoupling capacitors, pull-up and pull-down resistors, programming connections, and MOSFET switching into a functioning microcontroller-based system
- Technical documentation available here and code available here